# Semantics and Equational Axiomatisation for Quantum Communication: Presentation Proposal

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## 1 Overview of the Presentation

***What.*** We systematically study the semantics of quantum programs extended with classically controlled quantum communication: sending and receiving qubits, possibly depending on a classical measurement outcome. For example, the following is naturally expressed in imperative syntax, but not in the standard quantum circuit model.

```
p: qbit = in(); q: qbit = new(|0>);
if (measure(p) == |1>) { out(q); }
```

The end goal would be to develop an axiomatisation of program equivalence between communicating quantum programs.

***Why.*** Quantum communication is an important programming primitive in distributive quantum computing [3]. Textbook quantum protocols such as teleportation [2] and BB84 [1] already implicitly use communication primitives. Prior work already treats quantum communication within quantum process calculi [5, 6, 9, 15], where communication is entangled with the additional complexity of concurrency.

***How.*** We study the individual operations of quantum communication *in isolation*, as algebraic effects, following the tradition of Plotkin and Power [11]. We do so along three axes, following the triangle summarised by fig. 1. We propose a new algebraic theory for quantum communication building on [13], and study an operational and a denotational model of the theory. Finally, we show that denotational and operational equalities coincide.

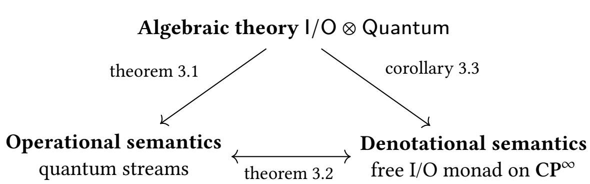



**Figure 1.** Three semantics of quantum communication and how they relate.

*This presentation is based on my MSc thesis [14] with Sam Staton (Oxford).

## 2 Background: Axiomatising Quantum Computing

We already know how to build the same triangle for qubit quantum computing without communication. Our starting point is [13]'s axiomatisation for qubit quantum computing, Quantum, based on *parameterised* algebraic theories (PAT). PATs are a generic framework for axiomatising the behaviour of computations dependending on certain resources in terms of operations and equations. Here, resources will correspond to qubits.

***Operations*** $\mathbf{op} : (p \mid m_1 ... m_k)$ represent computations which take $p$ qubits, classically choose one of the $k$ branches, say, $i$, and return $m_i$ qubits. Quantum has three operations: state preparation $\mathsf{new} : (0 \mid 1)$, unitary evolution $\mathsf{apply}_U : (n \mid n)$ for each $2^n \times 2^n$ unitary $U$, and classical control $\mathsf{measure} : (1 \mid 0, 0)$.

***Terms*** generated from the operations, follow the same intuition: a term $x_1 : m_1 ... x_k : m_k \mid q_1 ... q_p \vdash c$ takes $p$ input qubits $q_1 ... q_p$ and calls one of the $k$ continuations $x_1 ... x_k$ with the corresponding number of qubits. We can thus write for example the program

$$x : 1, y : 1 \mid q_1, q_2 \vdash \mathsf{measure}(q_1, x(q_2), y(q_2))$$

which measures a $q_1$ and continues with $x$ or $y$ with $q_2$ based on the measurement outcome.

***Equations*** are given in the form $\Gamma \mid \Delta \vdash c = c'$. Quantum for example specifies effect of measuring a $|0\rangle$ qubit using the following equation:

$$x : 0, y : 0 \mid \cdot \vdash \mathsf{new}(q.\mathsf{measure}(q, x, y)) = x$$

***Models: Denotational and Operational*** [13] proceeds to define a denotational model for the Quantum theory, where each term $x_1 : m_1 ... x_k : m_k \mid q_1 ... q_p \vdash c$ is interpreted as a complete positive unital (CPU) map

$$[\![c]\!] : \bigoplus_{i=1...k} \mathcal{M}_i(\mathbb{C}) \to \mathcal{M}_p(\mathbb{C}).$$

Remarkably, this model is *fully complete*: every CPU map of this type is the interpretation of some term $c$, and two CPU maps are equal iff their term representation are derivably equal in the equational theory.

We can in fact give an equivalent operational model for Quantum. Briefly, configurations would look like $\langle \rho, c\rangle$ where $\rho$ is the program state, $c$ the running program, and where transitions $\rightarrow_p$ occur probabilistically. We can then formulate in terms of $\rightarrow_p$ an operational notion of program equivalence $\simeq_{qu}$ equivalent with denotational equality.

# 3 Semantics for Quantum Communication

We proceed to extend our language with communication primitives, and strive to recover the same triangle.

## 3.1 An Algebraic Theory for Quantum Communication

Extending Quantum with two operations: $\mathsf{in} : (0 \mid 1)$ and $\mathsf{out} : (1 \mid 0)$ lets us represent classically controlled communication. We can express the example program as $x : 0 \mid \cdot \vdash \mathsf{in}(p.\mathsf{new}(q.\mathsf{measure}(p, x, \mathsf{out}(q, x))))$. We further add equations to make in and out commute with every Quantum operation. What we now obtain is $\mathsf{I/O} \otimes \mathsf{Quantum}$, where $\otimes$ denotes the commutative combination of two theories. This choice is motivated by quantum theory. For instance,

$$\mathsf{measure}(a,\ \mathsf{out}(b, x),\ \mathsf{out}(b, y)) = \mathsf{out}(b,\ \mathsf{measure}(a, x, y))$$

is required to derive the equivalence between: preparing a Bell pair, sending one end and discarding the other end; and tossing a fair coin and sending the outcome as a qubit.

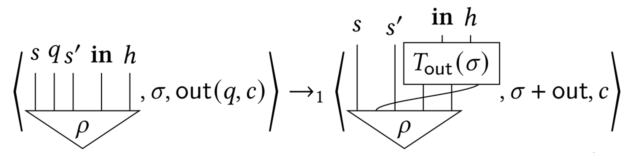
## 3.2 An Operational Model

The notion of communication that our theory soundly axiomatises can be demonstrated through an operational model. We work in, **CP** the category of finite dimensional C* algebras and complete positive maps, constructed as in [12].

The operational semantics is based on a *quantum stream*, specifying the environment our program communicates with. The idea is then to have a global quantum state $\rho : \mathbb{C} \to S \otimes \mathbf{qbit} \otimes H$, where $S$ is the program state, $H$ is the hidden stream state, and **qbit** is the next input qubit. Then, a stream is a state machine reacting to the program's communications: formally, a stream is specified by a hidden state type $H_\sigma$ and transition maps $T_{\mathsf{in}}(\sigma)$, $T_{\mathsf{out}}(\sigma)$ for every communication trace $\sigma \in \{\mathsf{in}, \mathsf{out}\}^*$.

We can then build an operational semantics with configuration $\langle \rho, \sigma, c\rangle$, where $\rho$ is the global state, $\sigma \in \{\mathsf{in}, \mathsf{out}\}^*$, $c$ is the program. Given a stream specification $t$ we can give a small step transition relation $\rightarrow_p^t$ defined inductively on the program. For example for the output operation, the rule is as in fig. 2: the output is performed, and the stream state is updated according to $t$. It is then not difficult to extend $\simeq_{qu}$ to communicating quantum programs, to obtain a sound model of $\mathsf{I/O} \otimes \mathsf{Quantum}$:

**Theorem 3.1.** $\Gamma \mid \Delta \vdash c = c' \implies \Gamma \mid \Delta \vdash c \simeq_{qu} c'$.




(out)

**Figure 2.** Operational Rule for out

## 3.3 A Denotational Model

***Strategy***

***The Quantum I/O Monad***

***Denotational Semantics***

**Theorem 3.2.** $\Gamma \mid \Delta \vdash c \simeq_{qu} c' \iff [\![c]\!] = [\![c']\!]$.

**Corollary 3.3.** $\Gamma \mid \Delta \vdash c = c' \implies [\![c]\!] = [\![c']\!]$.

# 4 Limitations and Future Work

We have thus obtained a sound axiomatisation of two notions of quantum communication, one operationally presented, one denotationally presented, which end up being equivalent. However, unlike the quantum computing case, we do not have completeness; it is unclear whether two programs $c$ and $c'$ being denotationally equal implies that they are derivably equal in $\mathsf{I/O} \otimes \mathsf{Quantum}$. Achieving this is our immediate next step.

# 5 Background: parameterised algebraic theories and QUANTUM

A parameterised algebraic theory (PAT) [13] is a signature of operations together with equations over their terms, whose operations carry *linear parameter contexts*: a typing judgement $\Gamma \mid \Delta \vdash c$ has $\Delta$ a list of parameter names (here, qubits) and $\Gamma$ a list of continuation variables with their valences. Operations $\mathbf{op} : (p \mid m_1 \ldots m_k)$ consume $p$ parameters and bind $m_i$ fresh parameters in each of $k$ continuations. Read as a type discipline, this is a linear type system for resource-sensitive effects.

Two combinators are standard: the *sum* (the free combination, no equations imposed between the two signatures) and the *tensor* (which adds the equation that every operation of one theory commutes with every operation of the other). Their constructions for the PATs of [13] are worked out in [14].

Staton's Quantum [13] is a PAT for qubit quantum computation with operations $\mathsf{new} : (0 \mid 1)$ (state preparation), $\mathsf{apply}_U : (n \mid n)$ (unitary evolution) and $\mathsf{measure} : (1 \mid 0, 0)$ (computational-basis measurement with classical control), together with quantum-theoretic equations such as the principle of deferred measurement. It is sound and complete with respect to its standard operator-algebra interpretation in **CP**, the finite-biproduct completion of the category **CPM** of matrix algebras and CP maps. The analogous picture to Figure 1 is settled in this I/O-free case. We extend it to communication.

## 6 The algebraic theory for I/O and QUANTUM

Extending Quantum with in : $(0 \mid 1)$ (receive a qubit) and out : $(1 \mid 0)$ (send a qubit) lets us write classically controlled communication directly. The theory $\mathsf{I/O} \otimes \mathsf{Quantum}$ is the *tensor* (not sum) of Quantum with a standard I/O theory [7]. There are two design points. First, in and out do not commute with each other — the order of I/O events is part of the observable behaviour of a program. Second, in and out do commute with all of Quantum's operations; this is motivated by quantum theory, since for instance

$$\mathsf{measure}(a,\ \mathsf{out}(b,x),\ \mathsf{out}(b,y)) = \mathsf{out}(b,\ \mathsf{measure}(a,x,y))$$

is needed to make a familiar quantum equivalence — preparing a Bell pair, sending one qubit and discarding the other equals tossing a fair coin and preparing the outcome — provable in the theory ([14], eq. 4.1.4). Tensors of theories can collapse (the Eckmann–Hilton argument for unital binary operations is the textbook example); whether $\mathsf{I/O} \otimes \mathsf{Quantum}$ collapses is ruled out *a posteriori* by the two models in §§4–5.

## 7 Operational semantics: quantum streams

A *quantum stream* is a $\{\mathsf{in},\mathsf{out}\}^*$-indexed family $q : \{\mathsf{in},\mathsf{out}\}^* \to \mathbf{Ob}\,\mathbf{CP}$ together with CPTP transitions $T(\sigma,\mathsf{in})$ and $T(\sigma,\mathsf{out})$ that update a hidden environment state at each I/O event. Streams are coalgebraic in flavour; we comment on a coalgebraic recasting in §7. A configuration $\langle \rho, \sigma, c\rangle$ pairs a normalised global state $\rho$ (over program qubits, communication qubit, and hidden state), an I/O trace $\sigma$, and a program $c$. The small-step probabilistic transition $\to_p^t$ is defined inductively in the program structure: measurement reduces probabilistically; other operations reduce deterministically; a continuation call terminates.

The induced *quantum equivalence* $c_1 \simeq_{qu} c_2$ holds when, for every stream and every initial state, the two programs yield the same probability-weighted distribution over (final state, continuation called, final I/O trace) triples. Terms of $\mathsf{I/O} \otimes \mathsf{Quantum}$ modulo $\simeq_{qu}$ form a model of $\mathsf{I/O}\otimes\mathsf{Quantum}$ ([14], Thm. 4.3.25): the equations of the PAT are validated by the stream dynamics.

## 8 Denotational semantics: a free I/O monad

The operational semantics lives in **CP**, which has only finite biproducts; the denotational sum over all I/O traces below needs infinitely many. Following [10], $\mathbf{CP}^\infty$ is the infinite-biproduct, dcpo-enriched completion of **CPM**, inheriting compact closure from **CPM**. On $\mathbf{CP}^\infty$ we define

$$T(X) \triangleq \mu Y.\ (\mathbf{qbit}^{\mathsf{in}}\otimes Y)\oplus(\mathbf{qbit}^{\mathsf{out}}\otimes Y)\oplus X \quad\cong \bigoplus_{\sigma\in\{\mathsf{in},\mathsf{out}\}^*} \mathsf{Aux}(\sigma)\otimes X,$$

where $\mathsf{Aux}(\epsilon) = \mathbb{C}$ and, for $s \in \{\mathsf{in},\mathsf{out}\}$, $\mathsf{Aux}(s\cdot\sigma) = \mathbf{qbit}^s \otimes \mathsf{Aux}(\sigma)$; the superscripts in/out are bookkeeping, both $\mathbf{qbit}^{\mathsf{in}}$ and $\mathbf{qbit}^{\mathsf{out}}$ are just **qbit**. For every I/O trace $\sigma$, the corresponding summand tensors the qubits exchanged along $\sigma$ with a value of type $X$.

The input case deserves a comment. An input operation morally has type $\mathbf{qbit} \multimap Y$. Compact closure of $\mathbf{CP}^\infty$ gives $\mathbf{qbit} \multimap Y \cong \mathbf{qbit}^* \otimes Y$, and the further fact that **qbit** is self-dual in **CPM** gives $\mathbf{qbit}^* \cong \mathbf{qbit}$, so the input summand can be written as $\mathbf{qbit}^{\mathsf{in}} \otimes Y$. The two readings are reconciled by a single string-diagrammatic move: the unnormalised maximally entangled state acts as a cup, bending the input wire.

Interpreting parameter contexts as tensor powers of **qbit** and continuation contexts as direct sums, each term $\Gamma \mid \Delta \vdash c$ denotes a Kleisli morphism

$$[\![c]\!]\ :\ [\![\Delta]\!] \longrightarrow T[\![\Gamma]\!] \ \cong \bigoplus_{\sigma\in\{\mathsf{in},\mathsf{out}\}^*,\ (x:n)\in\Gamma} \mathsf{Aux}(\sigma)\otimes\mathbf{qbit}^{\otimes n},$$

a family of CP maps indexed by (final I/O trace, continuation called) — the same indexing already used by $\simeq_{qu}$. The object $\mathbb{C}$ is a model of $\mathsf{I/O}\otimes\mathsf{Quantum}$ in $\mathsf{Kl}_T^{\mathrm{op}}$ ([14], Cor. 5.4.2).

## 9 Adequacy and full abstraction

The two concrete models coincide on program equivalence.

**Theorem 9.1** ([14], Thm. 5.3.4 (adequacy) and Thm. 5.3.6 (full abstraction))**.** *For all* $\Gamma \mid \Delta \vdash c_1, c_2$*:* $[\![c_1]\!] = [\![c_2]\!]$ *iff* $c_1 \simeq_{qu} c_2$.

This is the kind of abstract–concrete correspondence the workshop addresses: both sides are independent concrete techniques, and Theorem 9.1 shows they agree, with the abstract theory $\mathsf{I/O}\otimes\mathsf{Quantum}$ as the common interface that each refines. The proof goes via a *sum-of-paths* characterisation matching the operational sum over reduction sequences to the categorical sum over biproduct components, term by term. The matching of Theorem 9.1 does not, however, settle *completeness* of $\mathsf{I/O}\otimes\mathsf{Quantum}$: whether the equational theory derives every identification validated by the two concrete models is open ([14], Conjecture 5.4.4).

## 10 Discussion and open directions

The pattern *algebraic theory + two concrete models + adequacy/FA* echoes themes in the algebraic-effects programme of Plotkin and Power [11] and is instantiated, for Quantum alone, in [13]. The quantum I/O case is technically non-trivial because compact closure, infinite biproducts, and the possibility of a tensor-of-theories collapse each constrain the construction.

We would welcome discussion at the workshop on three open directions. *(a)* **Completeness** of $\mathsf{I/O} \otimes \mathsf{Quantum}$ for $\simeq_{qu}$: what would a suitable axiomatic logic look like for a matching like Theorem 9.1? *(b)* **Coalgebraic reading.** The stream of §4 is coalgebraic, and Ceragioli et al.'s coalgebraic model of quantum bisimulation [4] appears to provide a coalgebraic framing of related phenomena; we are also curious how our denotational semantics relates to the $\mathsf{Caus}$ construction of Kissinger–Uijlen [8]. *(c)* **Algorithmic verification.** The denotational semantics is probabilistic and indexed by I/O traces; is there a decidable fragment, or a probabilistic / quantum Hennessy–Milner logic against which equivalence in $\mathsf{I/O} \otimes \mathsf{Quantum}$ could be model-checked?

## References


[1] Charles H. Bennett and Gilles Brassard. 2014. Quantum cryptography: Public key distribution and coin tossing. *Theoretical Computer Science* 560 (2014), 7–11. https://doi.org/10.1016/j.tcs.2014.05.025 Theoretical Aspects of Quantum Cryptography – celebrating 30 years of BB84.

[2] Charles H. Bennett, Gilles Brassard, Claude Crépeau, Richard Jozsa, Asher Peres, and William K. Wootters. 1993. Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels. *Phys. Rev. Lett.* 70 (Mar 1993), 1895–1899. Issue 13. https://doi.org/10.1103/PhysRevLett.70.1895

[3] Marcello Caleffi, Michele Amoretti, Davide Ferrari, Jessica Illiano, Antonio Manzalini, and Angela Sara Cacciapuoti. 2024. Distributed quantum computing: A survey. *Computer Networks* 254 (Dec. 2024), 110672. https://doi.org/10.1016/j.comnet.2024.110672

[4] Lorenzo Ceragioli, Elena Di Lavore, Giuseppe Lomurno, and Gabriele Tedeschi. [n. d.]. A Coalgebraic Model of Quantum Bisimulation. ([n. d.]).

[5] Lorenzo Ceragioli, Fabio Gadducci, Giuseppe Lomurno, and Gabriele Tedeschi. 2024. Quantum Bisimilarity via Barbs and Contexts: Curbing the Power of Non-deterministic Observers. *Proc. ACM Program. Lang.* 8, POPL, Article 43 (jan 2024), 29 pages. https://doi.org/10.1145/3632885

[6] Simon Gay and Rajagopal Nagarajan. 2004. Communicating Quantum Processes. arXiv:quant-ph/0409052 [quant-ph] https://arxiv.org/abs/quant-ph/0409052

[7] Martin Hyland, Gordon Plotkin, and John Power. 2006. Combining effects: Sum and tensor. *Theoretical Computer Science* 357, 1 (2006), 70–99. https://doi.org/10.1016/j.tcs.2006.03.013 Clifford Lectures and the Mathematical Foundations of Programming Semantics.

[8] Aleks Kissinger and Sander Uijlen. 2019. A categorical semantics for causal structure. *Logical Methods in Computer Science* 15 (2019).

[9] Marie Lalire and Philippe Jorrand. 2004. A Process Algebraic Approach to Concurrent and Distributed Quantum Computation: Operational Semantics. arXiv:quant-ph/0407005 [quant-ph] https://arxiv.org/abs/quant-ph/0407005

[10] Michele Pagani, Peter Selinger, and Benoît Valiron. 2013. Applying quantitative semantics to higher-order quantum computing. *CoRR* abs/1311.2290 (2013). arXiv:1311.2290 http://arxiv.org/abs/1311.2290

[11] Gordon Plotkin and John Power. 2002. Notions of Computation Determine Monads. In *Foundations of Software Science and Computation Structures*, Mogens Nielsen and Uffe Engberg (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 342–356.

[12] Peter Selinger. 2007. Dagger Compact Closed Categories and Completely Positive Maps: (Extended Abstract). *Electronic Notes in Theoretical Computer Science* 170 (2007), 139–163. https://doi.org/10.1016/j.entcs.2006.12.018 Proceedings of the 3rd International Workshop on Quantum Programming Languages (QPL 2005).

[13] Sam Staton. 2015. Algebraic Effects, Linearity, and Quantum Programming Languages. In *Proceedings of the 42nd Annual ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages* (Mumbai, India) *(POPL '15)*. Association for Computing Machinery, New York, NY, USA, 395–406. https://doi.org/10.1145/2676726.2676999

[14] Théo Wang. 2024. *Algebraic Theories and Quantum Communication*. Master's thesis. University of Oxford, Oxford, United Kingdom. https://theo.wang/works/2024-09-MSc-thesis

[15] Mingsheng Ying, Yuan Feng, Runyao Duan, and Zhengfeng Ji. 2010. An Algebra of Quantum Processes. arXiv:0707.0330 [quant-ph] https://arxiv.org/abs/0707.0330